ROSHINI A

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PROFESSIONAL SUMMARY

Final-year B.Tech Artificial Intelligence & Machine Learning student seeking an entry-level AI / ML Engineer role. Hands-on experience building computer vision and machine learning solutions with Python, TensorFlow, MediaPipe and OpenCV, including real-time sign language translation, sensor-based emergency detection and K-Means clustering for academic analytics. Experience integrating ML into full-stack applications using React.js, Node.js, REST APIs and MongoDB.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript

AI / ML & Computer Vision: Machine Learning, CNN, K-Means Clustering, Computer Vision, Image Processing, TensorFlow, MediaPipe, OpenCV

Full-Stack & APIs: React.js, Node.js, Express.js, REST API Development, JWT Authentication, HTML5, CSS3, Bootstrap

Databases: MongoDB (NoSQL), MySQL (SQL)

Core Concepts: DBMS, Object-Oriented Programming (OOPS)

Tools: Git, GitHub, VS Code

PROJECTS

Real-Time Assistive Sign Language Translator | *MediaPipe, TensorFlow*

- Built a real-time assistive sign language translator using MediaPipe and TensorFlow to recognize and translate sign language gestures.

SafeSpace – Women Safety Android Application | *Android, Python, ML, GPS, SMS*

- Implemented sensor-based machine learning for automatic emergency detection, alongside one-touch alerts, real-time GPS tracking and SMS notifications.
- Added silent audio recording to the emergency workflow.

AI-Based Academic Performance Clustering System | *K-Means, React.js, Node.js, Express.js, MongoDB,*

- Developed and deployed a MERN-stack analytics platform that applies K-Means clustering to group students by academic performance and flag at-risk students.
- Built JWT-secured REST APIs in Node.js and Express.js and interactive React.js dashboards to present clustering results.

EDUCATION

B.Tech Artificial Intelligence & Machine Learning **2023 – 2027**
Bannari Amman Institute of Technology, Sathyamangalam | CGPA: 8

HSC – Vidyaa Vikas Matric Higher Secondary School, Karamadai | Aggregate: 90.5% **2021 – 2023**

CERTIFICATIONS

- Programming in Java – NPTEL (Score: 90%)
- ServiceNow Certified Application Developer (CAD)
- ServiceNow Certified System Administrator (CSA)

ACHIEVEMENTS & ADDITIONAL INFORMATION

Hackathons: Smart India Hackathon (SIH); Women Hackathon (TNWISE)

Soft Skills: Willingness to learn, fast learner

Languages: English, Tamil, French